Assignment-4

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B:05

a) 1) No. of machines = 3
machines = A, B, C

No. of plays = n play = 10

Initial values

$$N(A) = N(B) = N(C) = 0$$
$$R(A) = R(B) = R(C) = 0$$
$$Q(A) = Q(B) = Q(C) = 0$$

UCB:

$$\boxed{UCB(a) = Q(a) + \sqrt{\frac{2\ln(t)}{N(a)}}}$$

If $N(a) = 0$
$UCB(a) = \infty$

2) Incremented sample-Avg formula:

$$\boxed{Q_{new}(a) = Q_{old}(a) + \frac{R - Q_{old}(a)}{N(a)}}$$

Initial table

Machine	N	R	Q	UCB
A	0	0	0	∞
B	0	0	0	∞
C	0	0	0	∞

for play-1

$R = 3$

$N(A) = 0$

$N(A) = 0+1 = 1$

$R(A) = 3$

$Q(A) = 0 + \frac{3-0}{1}$

$\therefore \boxed{Q(A) = 3}$

After play 1:

Machine	N	P	Q
A	1	3	3
B	0	0	0
C	0	0	0

Reward $R = 6$

$N(B) = 0$

$N(B) = 1$

$R(B) = 6$

$Q(B) = 0 + \frac{6-0}{1} = 6 = Q(B)$

After play 2:

	N	P	Q
A	1	3	3
B	1	6	6
C	0	0	0

$R = 4$

$N(C) = 1$

$R(C) = 4$

$Q(C) = 0 + \frac{4-0}{1}$

$\boxed{Q(C) = 4}$

UCB calculation

$Q(A) = 3$ $\quad N(A) = 1$

$Q(B) = 6$ $\quad N(B) = 1$

$Q(C) = 4$ $\quad N(C) = 1$

for Machine A:

$UCB(A) = 3 + \sqrt{\frac{2\ln(4)}{1}}$

$\ln(4) \approx 1.3863$

$= 3 + \sqrt{2(1.3863}$

$= 3 + 1.665$

$\boxed{UCB(A) \approx 4.665}$

Machine B:

$UCB(B) = 6 + \sqrt{\frac{2\ln(4)}{1}}$

$= 6 + 1.665$

$\boxed{UCB(B) \approx 7.665}$

for Machine c:

$$UCB(c) = 4 + \sqrt{\frac{2\ln(4)}{1}}$$

$$\approx 4 + 1.665$$

$$UCB(c) \approx 5.665$$

$\therefore UCB(B) > UCB(C) > UCB(A)$ //

Hence, Machine B is selected //.

Incremented Q Calculation:

$Q(B) = 6$

$R = 7$

$N(B) = 2$

$$Q_{new}(B) = Q_{old}(B) + \frac{R - Q_{old}(B)}{N(B)}$$

$$Q_{new}(B) = 6 + \frac{7-6}{2} = 6 + \frac{1}{2}$$

$$\approx 6.5$$

A: 3, 4, 2, 5

B: 6, 7, 5, 8

C: 4, 3, 5, 4

Machine A = $Q(A) = \frac{3+4+2+5}{4} = 14/4 = 3.5$

Machine B = $Q(B) = \frac{6+7+5+8}{4} = \frac{26}{4} = 6.5$

Machine C = $Q(C) = \frac{4+3+5+4}{4} = 16/4 = 4$

$Q(B) > Q(C) > Q(A)$

$\therefore$ Total Rewards $= 14 + 26 + 16 \Rightarrow 56$ //.